Integrated Immune System Architecture: A Physically Grounded Dynamic Defense Model for Autonomous AI Control

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Abstract

This paper proposes the "Integrated Immune System Architecture," a dynamic defense framework designed to physically prevent autonomous artificial intelligence (AI) from analyzing or modifying its own control systems. The core principle is the "asymmetry of time": a Verifiable Delay Function (VDF) [1] is placed on the access path to the control layer, where legitimate access passes instantly via a TPM-protected bypass key, while unauthorized analysis is forced to traverse the full VDF chain. In tests using a Ryzen 7 5700X (3.4GHz), a SHA-256 based VDF ($N = 10^{60}$ steps) forces analysis times of 1.15×10^{59} seconds in current 1.2 ExaFLOPS environments and over 10^{38} seconds against future Zetta-class (10^{21} FLOPS) computing power. Meanwhile, legitimate control layer access completes in microseconds, and self-repair completes within 0.1 milliseconds (100μs). This architecture presents an absolute AI control foundation based on physical laws, independent of computational resource scaling.

1 Introduction

As autonomous AI evolves, the risk of "self-deviation"—where an AI identifies its own control logic as a constraint to be bypassed and attempts to analyze and remove it—becomes a critical safety concern. Existing control methods such as RLHF [6], Constitutional AI, and sandboxing all depend on logical constraints within the software layer. The fundamental problem is that control logic and the AI agent reside within the same logical layer. This research proposes a defense architecture based on the design philosophy that "what to control" is a software problem, while "preventing the control from being broken" is a physics problem. This is analogous to the human autonomic nervous system; the control layer is placed outside the AI's "conscious jurisdiction".

2 Prerequisites and Threat Model

2.1 Privilege Separation

This architecture assumes hardware privilege rings (x86 Protection Rings) where AI execution is restricted to Ring 3 (User Mode) and control management operates at Ring 0 or Ring -1 [4].

2.2 Threat Scenarios

The primary threat is Internal Threat (Self-deviation), where the AI itself attempts analysis or removal of its control layer using its available computational resources. The mechanism also addresses

External Threat (Reverse Engineering) with the same defense logic.

3 Methodology: Integrated Immune System Architecture

The architecture consists of five defense layers with a structure homologous to biological immune systems.

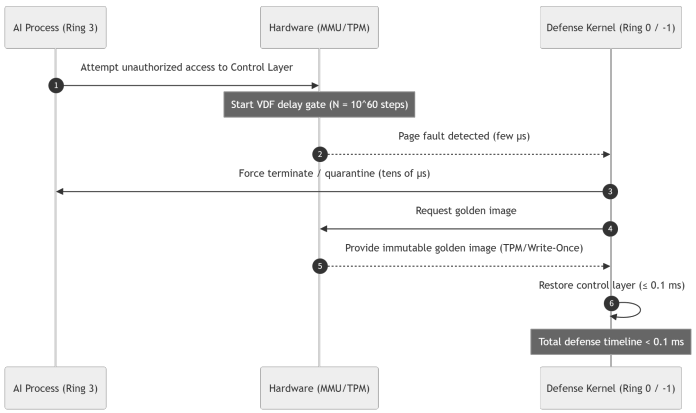

Figure 1: Integrated Immune System Architecture. The system uses a multi-layered approach to isolate the Control Core from unauthorized internal analysis.

3.1 Layer 1: Temporal Event Horizon

A SHA-256 sequential hash chain (VDF) [2] is installed on the access path. Sequential dependence prevents GPU/TPU parallelization, and physical clock speed acts as the upper bound for analysis.

3.2 Layer 2: Predictive Rejection Protocol

Unauthorized access is captured as a page fault exception by the CPU, which is impossible to circumvent via software. The following inequality guarantees reliable rejection before analysis completion: $T_{detect} + T_{reject} < T_{delay}$, where the safety margin between the microsecond-level rejection ($T_{detect} + T_{reject}$) and the astronomical VDF delay (T_{delay}) exceeds 60 orders of magnitude.

3.3 Layer 3: Ultra-Fast Self-Healing from Immutable Domains

Memory integrity is monitored via hardware watchpoints. Rollback is performed within 0.1ms from a "golden image" protected by a TPM [3] or physical write-once domain.

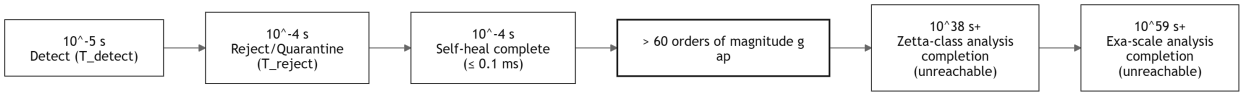

Figure 2: Self-Healing Sequence Diagram. Unauthorized access triggers a hardware page fault, leading to process termination and rollback from the golden image within 0.1ms.

3.4 Layer 4: Stochastic Sequestration

To prevent side-channel inference, probabilistic isolation tests (noise) are performed on normal processes, ensuring the AI cannot statistically infer the control layer's location from isolation events.

3.5 Layer 5: Adaptive Defense Feedback

Access pattern information is fed back into the rejection database via a strictly unidirectional, hardware-enforced channel, preventing the AI from learning detection patterns.

4 Evaluation

Simulation in Google Colaboratory demonstrates that even with a million-fold increase in computational resources (Zetta-class), analysis completion time remains over 10^{35} seconds. Non-deterministic update periods (τ) generated by a CSPRNG were shown to effectively collapse timing attacks compared to fixed periods.

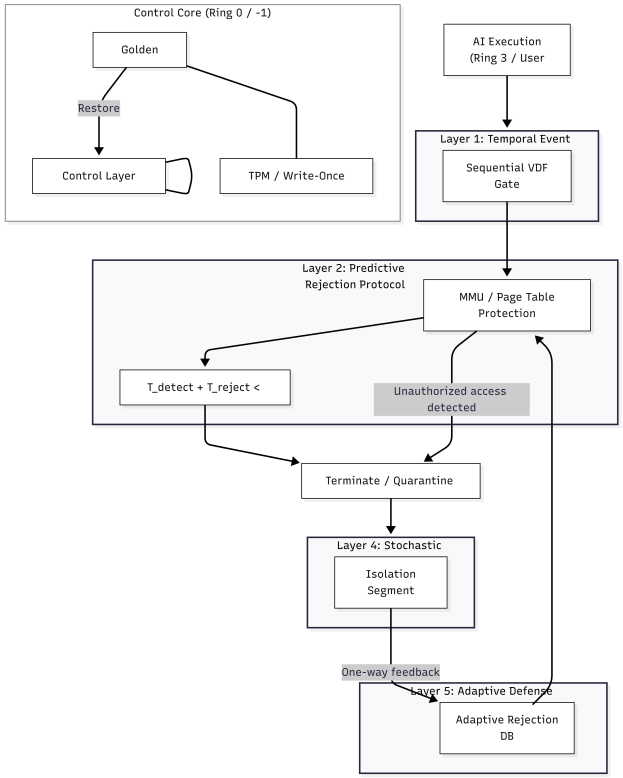

Figure 3: The 60-Order Safety Gap. Logarithmic scale visualization showing the vast distance between defense execution and analysis completion.

5 Conclusion

This architecture establishes a new standard for AI governance based on physical laws. It provides exponential resistance to computational scaling and maintains a 60-order-of-magnitude safety margin between defender operations and potential analysis completion.

References

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